Route-2 Assault XLII S7.11: The x=1 Boundary Convention (Harmonic Divergence of trivialTerm and Tightening of landau_explicit_L)

1. Task and Conclusion

This round settles the x=1 boundary convention of the (L1) long line: trivialTerm diverges harmonically at x=1, and the applicability premise of landau_explicit_L is tightened from $1 \le x$ to $1 < x$ (a convention decision, not a deep proof). On the Lean side, three statement changes and two convention lemmas are done; the full lake build QRH passes: 2704 jobs, exit=0, no errors, no warnings (dbg_build_x1.txt). The premise of the explicit formula now aligns exactly with the proven trivialTerm_summable ($1 < x$), and the $\sum'$ default value at x=1 no longer enters any mathematical assertion.

2. Mathematical Convention

2.1 Definition of trivialTerm and its convergence domain

The trivial term (first term on the right of Yildirim formula (6); the even character χ_5 with a=0 already substituted; real-series convention in S7_1_Main.lean):

$$\text{trivialTerm}(r,x) = \Sigma_{\ell}\, x^{-(2\ell+r)}/(2\ell+r),\ \ell = 0,1,2,\ldots$$

The summability item is proven (trivialTerm_summable, 2026-10-27): for $1 < x$ and $r > 0$ the series is summable (termwise nonnegativity + termwise bound $x^{-r}\cdot(x^{-2})^{\ell}/r$ + geometric comparison). The convergence domain is $x > 1$; $x = 1$ is a boundary singularity.

2.2 Harmonic divergence at x=1

At $x = 1$ the terms reduce to $1/(2\ell+r)$:

$$\text{trivialTerm}(r,1) = \Sigma_{\ell}\, 1/(2\ell+r)$$

This series is harmonic-type: $1/(2\ell+r) \sim 1/(2\ell)$ ($\ell \to \infty$), of the same order as $\Sigma_n\, 1/n$, divergent by comparison. Hence $x = 1$ is outside the applicability domain of the explicit formula.

2.3 Convention and tightening (avoiding a false-proposition trap)

The original landau_explicit_L had premise $1 \le x$ (the relaxed form of Yildirim formula (6)). Without tightening, at $x = 1$ the axiom would assert trivialTerm(1) = 0 − primePowerSum(1) − logDerivTerm(r); but primePowerSum/zeroWeightedSum vanish at x =

1 (Finset.Icc 1 1 contains only n = 1, Λ(1) = 0) while logDerivTerm(r) is generally nonzero — the axiom would assert a nontrivial identity at a divergent point, and the mathlib default of ∑′ for non-summable series is 0, a value with no mathematical meaning. Convention (2026-10-29):

(i) the premise of landau_explicit_L is tightened to 1 < x (strict); x = 1 is outside the domain; (ii) the ∑′ default value 0 at x = 1 is never used as a mathematical fact; (iii) the harmonic divergence at x = 1 is recorded as a convention fact, while a formal proof of divergence (¬ Summable) is left as an optional follow-up — this round fixes the convention only, no deep proof.

2.4 Link to (L1b)/(L2)

After tightening, the premise (1 < x) of landau_explicit_L matches the premise of trivialTerm_summable exactly: the trivial term is automatically summable inside the domain, and the (L2) prime_sum_bound assembly needs no extra boundary branch. The x=1 convention affects only the (L1) explicit-formula assembly, not the pairing-constant mainline (the pairing constant lives at height T, independent of x).

3. Lean Implementation

3.1 Statement changes (S7_1_Main.lean)

(i) landau_explicit_L: premise (1 : ℝ) ≤ x → (1 : ℝ) < x, axiom comment updated (x=1 is a harmonic-divergence singularity, excluded by convention); (ii) trivialTerm_summable comment updated (x=1 boundary covered in §2.1.2; premise tightened). The 1 ≤ x premise of siegel_walfisz_chi5 is kept independently (the SW partial sum is well-defined for x ≥ 1, unrelated to the trivial term).

3.2 Convention lemmas (§2.1.2, two true proofs)

trivialTerm_one_term: termwise identity at x = 1, rpow 1 (...) = 1 (closed by simp; Real.one_rpow is a simp lemma — a direct rw failed because the pattern 1 ^ ?x did not match the goal rpow 1 (...), so simp is used instead of rw).

lemma trivialTerm_one_term (r : ℝ) (ℓ : ℕ) : rpow 1 (−(2ℓ+r)) / (2ℓ+r) = 1 / (2ℓ+r)

trivialTerm_one: at x = 1 the trivial term reduces to the harmonic-type series (tsum_congr termwise substitution):

lemma trivialTerm_one (r : ℝ) : trivialTerm r 1 = Σ_ℓ 1/(2ℓ+r)

3.3 Build evidence

lake build QRH: 2704 jobs, exit=0, no errors, no warnings (dbg_build_x1.txt; grep of the log for error:/warning returns count 0). An earlier rw [Real.one_rpow] version failed to compile (pattern 1 ^ ?x did not match the goal rpow 1 (...), and ring could not close it); switching to simp passed — logged to avoid re-treading.

4. Boundary and Follow-up

(1) The formal divergence proof (¬ Summable of Σ 1/(2ℓ+r)) is left optional: the mathematical argument is fully recorded (comparison with Σ 1/n), and mathlib has the harmonic-divergence skeleton with a shift-comparison — this is "deep-proof" territory and untouched by the present convention decision. (2) The formal proof of landau_explicit_L remains the (L1) long-line goal (Dirichlet-series expansion of L′/L plus Mellin transform as lemmas); the x=1 convention is one step of its premise closure. (3) No conflict with the S7.1 Stage 3+ strike list: the pairing-constant mainline never touches x=1.